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Experiment No. 10

1. Dataset Source:

The dataset used in this experiment is the MNIST Dataset.

🔗 Source link:
https://yann.lecun.com/exdb/mnist/

2. Dataset Description

The MNIST dataset is a collection of handwritten digit images used for image processing and machine learning tasks.

a) **Key Features:**

- **Images:** Grayscale images of digits (0–9)
- **Image Size:** 28 × 28 pixels
- **Pixel Range:** 0–255 (normalized to 0–1)
- **Training Samples:** 60,000 images
- **Testing Samples:** 10,000 images

b) **Target Variable:**

- Digits (0 to 9) → **Not used in this experiment**
- This is because Autoencoder is an **unsupervised learning algorithm**

c) **Important Characteristics:**

- Simple and clean dataset
- Widely used for benchmarking

- Suitable for image reconstruction tasks

3. Mathematical Formulation of the Algorithm

Autoencoder consists of two parts:

a) Encoder:

Transforms input into compressed representation:

$$z = f(x) = \sigma(Wx + b)$$

b) Decoder:

Reconstructs original image:

$$\hat{x} = g(z) = \sigma(W'z + b')$$

c) Loss Function:

The model minimizes reconstruction error:

$$L = -\sum [x \log(\hat{x}) + (1 - x) \log(1 - \hat{x})]$$

In your code:

- Loss used = Binary Cross Entropy
- Goal = Minimize difference between original and reconstructed image

4. Algorithm Limitations

Autoencoders have some limitations:

- Cannot generate completely new data like GANs
- Performance depends on architecture design
- May fail on very noisy or complex datasets
- Overfitting can occur if model is too deep
- Not suitable for classification directly.

5. Methodology / Workflow

Step-by-step Process:

1. **Load Dataset**
    - MNIST dataset is loaded from Keras
2. **Data Preprocessing**
    - Normalize pixel values (0–1)
    - Reshape images to (28, 28, 1)
3. **Add Noise**
    - Gaussian noise added to images
    - Creates noisy input for training

4. **Build Autoencoder**
    - Encoder → Conv + Pooling
    - Decoder → Upsampling + Conv
5. **Model Training**
    - Input: Noisy images
    - Output: Clean images
6. **Prediction**
    - Model reconstructs denoised images
7. **Visualization**
    - Original images
    - Noisy images
    - Denoised output

6. Performance Analysis

a) Evaluation Metric:

- Loss Function: Binary Cross Entropy

b) Observations:

- Training loss decreases over epochs
- Denoised images are clearer than noisy images
- Model successfully removes Gaussian noise

c) Interpretation:

- Autoencoder learns important features of images
- It reconstructs images by removing noise patterns

7. Hyperparameter Tuning

a) Parameters Used:

- Epochs = 5
- Batch Size = 128
- Noise Factor = 0.5
- Optimizer = Adam.

🔹 Tuning Process:

Parameter	Effect
Epochs ↑	Better learning but more time
Noise Factor ↑	Harder denoising task
Filters ↑	Better feature extraction
Batch Size ↓	Slower but more accurate

b) Impact:

- Increasing epochs improves reconstruction quality
- Proper noise level ensures effective learning
- Balanced architecture avoids overfitting

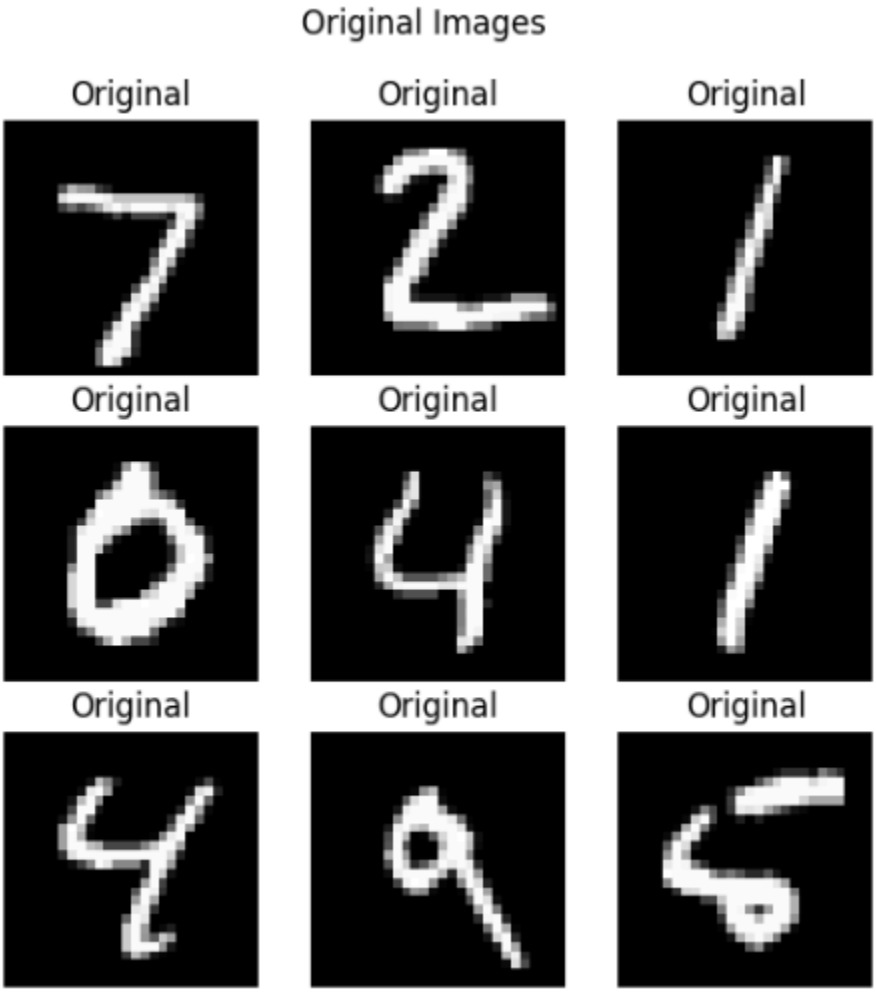
CODE:

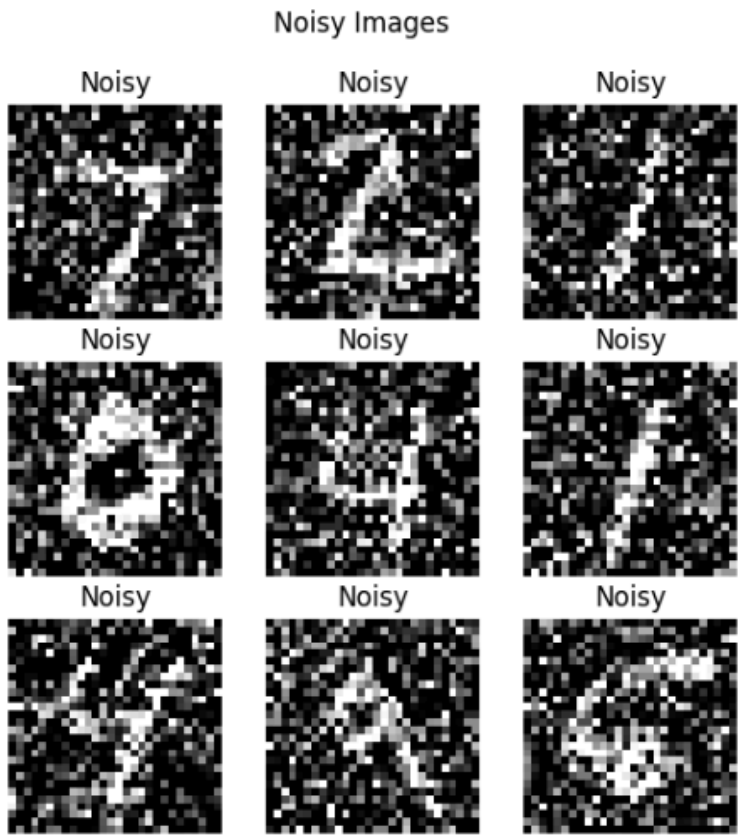
```
# 1. Import Libraries
import numpy as np
import matplotlib.pyplot as plt

from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Input, Conv2D, MaxPooling2D, UpSampling2D

(x_train, _), (x_test, _) = mnist.load_data()

# Normalize
x_train = x_train.astype('float32') / 255.
x_test  = x_test.astype('float32') / 255.

# Reshape
x_train = x_train.reshape(-1, 28, 28, 1)
x_test  = x_test.reshape(-1, 28, 28, 1)

print("Dataset Loaded Successfully")

noise_factor = 0.5

x_train_noisy = x_train + noise_factor * np.random.normal(size=x_train.shape)
x_test_noisy  = x_test + noise_factor * np.random.normal(size=x_test.shape)

# Clip values
x_train_noisy = np.clip(x_train_noisy, 0., 1.)
x_test_noisy  = np.clip(x_test_noisy, 0., 1.)

print("Noise Added")
```

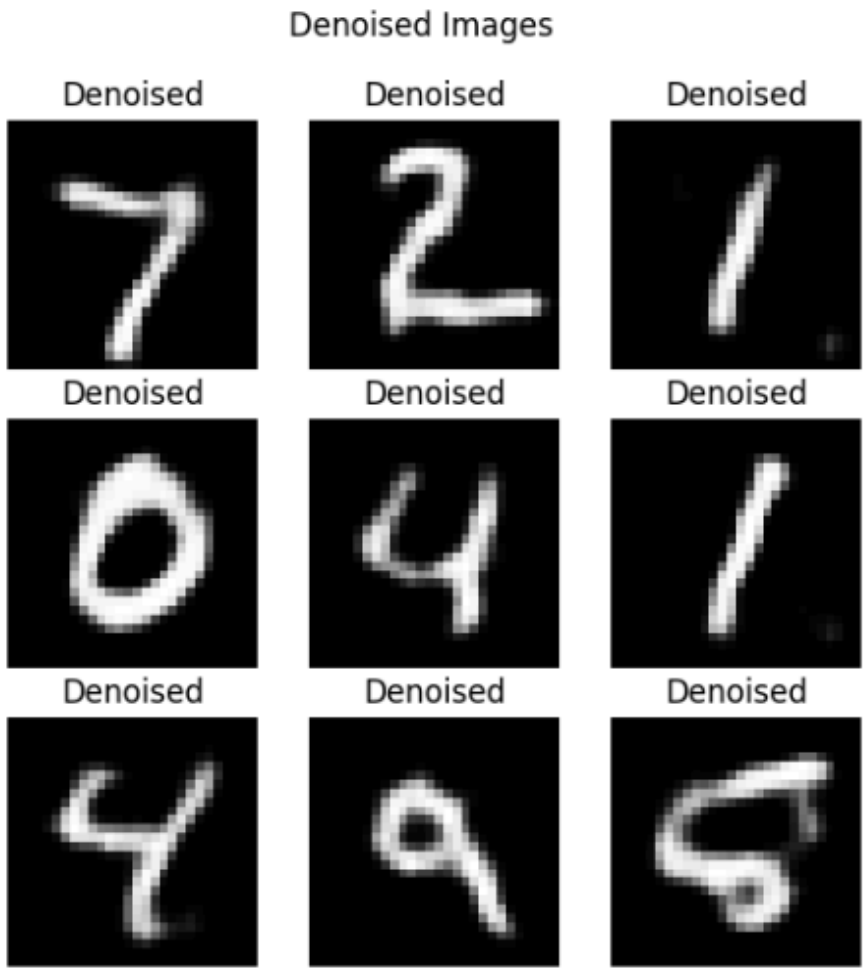
```
input_img = Input(shape=(28,28,1))

# Encoder
x = Conv2D(32, (3,3), activation='relu', padding='same')(input_img)
x = MaxPooling2D((2,2), padding='same')(x)

x = Conv2D(16, (3,3), activation='relu', padding='same')(x)
encoded = MaxPooling2D((2,2), padding='same')(x)

# Decoder
x = Conv2D(16, (3,3), activation='relu', padding='same')(encoded)
x = UpSampling2D((2,2))(x)

x = Conv2D(32, (3,3), activation='relu', padding='same')(x)
x = UpSampling2D((2,2))(x)

decoded = Conv2D(1, (3,3), activation='sigmoid', padding='same')(x)

# Model
autoencoder = Model(input_img, decoded)

# Compile
autoencoder.compile(optimizer='adam', loss='binary_crossentropy')

print("Model Built Successfully")

history = autoencoder.fit(
    x_train_noisy, x_train,
    epochs=5,
    batch_size=128,
    shuffle=True,
    validation_data=(x_test_noisy, x_test)
)
```

```
print("Training Completed")

decoded_imgs = autoencoder.predict(x_test_noisy)

print("Prediction Done")

plt.figure(figsize=(6,6))
for i in range(9):
    plt.subplot(3,3,i+1)
    plt.imshow(x_test[i].reshape(28,28), cmap='gray')
    plt.title("Original")
    plt.axis('off')
plt.suptitle("Original Images")
plt.show()
plt.figure(figsize=(6,6))
for i in range(9):
    plt.subplot(3,3,i+1)
    plt.imshow(x_test_noisy[i].reshape(28,28), cmap='gray')
    plt.title("Noisy")
    plt.axis('off')
plt.suptitle("Noisy Images")
plt.show()
plt.figure(figsize=(6,6))
for i in range(9):
    plt.subplot(3,3,i+1)
    plt.imshow(decoded_imgs[i].reshape(28,28), cmap='gray')
    plt.title("Denoised")
    plt.axis('off')
plt.suptitle("Denoised Images")
plt.show()
```

OUTPUT:


Original Images
Original
Original
Original
Original
Original
Original
Original
Original
Original


Noisy Images
Noisy
Noisy
Noisy
Noisy
Noisy
Noisy
Noisy
Noisy
Noisy


Denoised Images
Denoised
Denoised
Denoised
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